

PAST EXAMINATION QUESTIONS (FP1)

TOPIC 4: POLAR COORDINATES

1. The curves C_1 and C_2 have polar equations, for $0 \leq \theta \leq \frac{1}{2}\pi$, as follows:

$$C_1: r = 2(e^\theta + e^{-\theta}),$$

$$C_2: r = e^{2\theta} - e^{-2\theta}.$$

The curves intersect at the point P where $\theta = \alpha$.

- (i) Show that $e^{2\alpha} - 2e^\alpha - 1 = 0$. Hence find the exact value of α and show that the value of r at P is $4\sqrt{2}$. [6]
- (ii) Sketch C_1 and C_2 on the same diagram. [3]
- (ii) Find the area of the region enclosed by C_1 , C_2 and the initial line, giving your answer correct to 3 significant figures. (5.82) [5]

Oct/Nov 2019/11/12/13/11O

2. The curve C_1 has polar equation $r^2 = 2\theta$, for $0 \leq \theta \leq \frac{1}{2}\pi$.

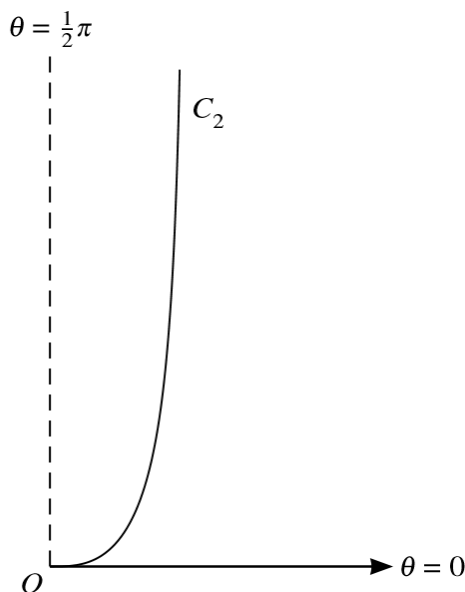
- (i) The point on C_1 furthest from the line $\theta = \frac{1}{2}\pi$ is denoted by P . Show that, at P ,

$$2\theta \tan \theta = 1$$

and verify that this equation has a root between 0.6 and 0.7. [5]

The curve C_2 has polar equation $r^2 = \theta \sec^2 \theta$, for $0 \leq \theta < \frac{1}{2}\pi$. The curves C_1 and C_2 intersect at the pole, denoted by O , and at another point Q .

- (ii) Find the exact value of θ at Q . $(\frac{\pi}{4})$ [2]
- (iii) The diagram below shows the curve C_2 . Sketch C_1 on this diagram. [2]



- (iv) Find, in exact form, the area of the region OPQ enclosed by C_1 and C_2 . $\left(\frac{1}{4}\ln 2 + \frac{\pi}{8}\left(\frac{\pi}{4} - 1\right)\right)$ [5]

May/June 2019/11/12/11E

3. The curve C has polar equation $r^2 = \ln(1 + \theta)$, for $0 \leq \theta \leq 2\pi$.
 (i) Sketch C . [2]
 (ii) Using the substitution $u = 1 + \theta$, or otherwise, find the area of the region bounded by C and the initial line, leaving your answer in an exact form. $\left(\left(\pi + \frac{1}{2}\right)\ln(2\pi + 1) - \pi\right)$ [5]

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4. The curve C has polar equation

$$r = 5\sqrt{\cot \theta}$$

where $0.01 \leq \theta \leq \frac{1}{2}\pi$.

- (i) Find the area of the finite region bounded by C and the line $\theta = 0.01$, showing full working. Give your answer correct to 1 decimal place. (57.6) [3]
 Let P be the point C where $\theta = 0.01$.
 (ii) Find the distance of P from the initial line, giving your answer correct to 1 decimal place. (0.5) [2]
 (iii) Find the maximum distance of C from the initial line. $\left(\frac{5\sqrt{2}}{2}\right)$ [3]
 (iv) Sketch C . [2]

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5. The curve C has polar equation $r = a \cos 3\theta$, for $-\frac{1}{6}\pi \leq \theta \leq \frac{1}{6}\pi$, where a is a positive constant.
 (i) Sketch C . [2]
 (ii) Find the area of the region enclosed by C , showing full working. $\left(\frac{\pi a^2}{12}\right)$ [3]
 (iii) Using the identity $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$, find a cartesian equation of C . $\left((x^2 + y^2)^2 = ax(x^2 - 3y^2)\right)$ [3]

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6. The curve C has polar equation $r = \cos 2\theta$, for $-\frac{1}{4}\pi \leq \theta \leq \frac{1}{4}\pi$.
 (i) Sketch C . [2]
 (ii) Find the area of the region enclosed by C , showing full working. $\left(\frac{\pi}{8}\right)$ [3]
 (iii) Find a cartesian equation of C . $\left((x^2 + y^2)^{\frac{3}{2}} = x^2 - y^2\right)$ [3]

May/June 2018/11/12/3

7. The curves C_1 and C_2 have polar equations, for $0 \leq \theta \leq \pi$, as follows:

$$\begin{aligned} C_1: r &= a, \\ C_2: r &= 2a |\cos \theta|, \end{aligned}$$

where a is a positive constant. The curves intersect at the points P_1 and P_2 .

- (i) Find the polar coordinates of P_1 and P_2 . $\left(\left(a, \frac{\pi}{3}\right), \left(a, \frac{2\pi}{3}\right)\right)$ [2]
 (ii) In a single diagram, sketch C_1 , C_2 and their line of symmetry. [3]
 (iii) The region R enclosed by C_1 and C_2 is bounded by the arcs OP_1 , P_1P_2 and P_2O , where O is the pole. Find the area of R , giving your answer in exact form. $\left(-\frac{\pi a^2}{6} + \frac{a^2\sqrt{3}}{2}\right)$ [5]
May/June 2018/13/8

8. The polar equation of a curve C is $r = a(1 + \cos \theta)$ for $0 \leq \theta < 2\pi$, where a is a positive constant.
 (i) Sketch C . [2]
 (ii) Show that the cartesian equation of C is $x^2 + y^2 = a\left(x + \sqrt{x^2 + y^2}\right)$. [2]
 (iii) Find the area of the sector of C between $\theta = 0$ and $\theta = \frac{1}{3}\pi$. $\left(\frac{a^2}{16}(4\pi + 9\sqrt{3})\right)$ [4]
Oct/Nov 2017/11/12/13/110

9. The curve C has polar equation $r = a(1 + \sin \theta)$ for $-\pi < \theta \leq \pi$, where a is a positive constant.
 (i) Sketch C . [2]
 (ii) Find the area of the region enclosed by C . $\left(\frac{3\pi a^2}{2}\right)$ [4]
May/June 2017/11/12/11 (part)

10. A curve C has polar equation $r = 2a \cos\left(2\theta + \frac{1}{2}\pi\right)$ for $0 \leq \theta < 2\pi$, where a is a positive constant.
 (i) Show that $r = -2a \sin 2\theta$ and sketch C . [4]
 (ii) Deduce that the cartesian equation of C is $(x^2 + y^2)^{\frac{3}{2}} = -4axy$. [2]
 (iii) Find the area of one loop of C . $\left(\frac{\pi a^2}{2}\right)$ [5]
 (iv) Show that, at the points (other than the pole) at which a tangent to C is parallel to the initial line, $2 \tan \theta = -\tan 2\theta$. [3]
May/June 2017/13/11E

11. A curve C has polar equation $r^2 = 8 \operatorname{cosec} 2\theta$ for $0 < \theta < \frac{1}{2}\pi$. Find a cartesian equation of C . ($xy = 4$) [3]
 Sketch C . [2]

Determine the exact area of the sector bounded by the arc of C between $\theta = \frac{1}{6}\pi$ and $\theta = \frac{1}{3}\pi$, the half line $\theta = \frac{1}{6}\pi$ and the half-line $\theta = \frac{1}{3}\pi$. (**2 ln 3**) [3]

[It is given that $\int \operatorname{cosec} x \, dx = \ln \left| \tan \frac{1}{2}x \right| + c$.]

May/June 2016/11/12/4

12. A curve has polar equation $r = \frac{1}{1 - \cos \theta}$, for $0 < \theta < 2\pi$. Find, in the form $y^2 = f(x)$, the cartesian equation of the curve. ($y^2 = 1 + 2x$) [3]

Hence, sketch the curve, and shade the region whose area is given by

$$\frac{1}{2} \int_{\frac{1}{2}\pi}^{\frac{3}{2}\pi} \frac{1}{(1 - \cos \theta)^2} d\theta. \quad [3]$$

Using the cartesian equation of the curve, find the area of this region. $\left(\frac{2}{3}\right)$ [3]

May/June 2016/13/7

13. The curves C_1 and C_2 have polar equations

$$C_1: r = \frac{1}{\sqrt{2}}, \text{ for } 0 \leq \theta < 2\pi,$$

$$C_2: r = \sqrt{\sin \frac{1}{2}\theta}, \text{ for } 0 \leq \theta \leq \pi.$$

Find the polar coordinates of the point of intersection of C_1 and C_2 . $\left(\frac{1}{\sqrt{2}}, \frac{1}{3}\pi\right)$ [2]

Sketch C_1 and C_2 on the same diagram. [3]

Find the exact value of the area of the region enclosed by C_1, C_2 and the half-line $\theta = 0$. $\left(\frac{1}{12}\pi + \frac{\sqrt{3}}{2} - 1\right)$ [4]

May/June 2015/11/12/5

14. A circle has polar equation $r = a$, for $0 \leq \theta < 2\pi$, and a cardioid has polar equation $r = a(1 - \cos \theta)$, for $0 \leq \theta < 2\pi$, where a is a positive constant. Draw sketches of the circle and the cardioid on the same diagram. [3]

Write down the polar coordinates of the points of intersection of the circle and the cardioid. $\left((a, \frac{\pi}{2}), (a, \frac{3\pi}{2})\right)$ [2]

Show that the area of the region that is both inside the circle and inside the cardioid is [6]

$$\left(\frac{5}{4}\pi - 2\right)a^2.$$

Oct/Nov 2014/11/12/13/8

15. The curve C has polar equation $r = a(1 + \sin \theta)$, where a is a positive constant and $0 \leq \theta < 2\pi$. Draw a sketch of C . [2]

Find the exact value of the area of the region enclosed by C and the half-lines $\theta = \frac{1}{3}\pi$ and $\theta = \frac{2}{3}\pi$. $\left(a^2 \left(\frac{1}{4}\pi + 1 + \frac{1}{8}\sqrt{3}\right)\right)$ [4]

May/June 2014/11/12/5

16. The curve C has cartesian equation $(x^2 + y^2)^2 = 2a^2xy$, where a is a positive constant. Show that the polar equation of C is $r^2 = a^2 \sin 2\theta$. [3]

Sketch C for $-\pi < \theta < \pi$. [2]

Find the area enclosed by one loop of C . $\left(\frac{a^2}{2}\right)$ [2]

May/June 2014/13/4

17. The curve C has polar equation $r = 2e^\theta$, for $\frac{1}{6}\pi \leq \theta \leq \frac{1}{2}\pi$. Find
(i) the area of the region bounded by the half-lines $\theta = \frac{1}{6}\pi, \theta = \frac{1}{2}\pi$ and C , $\left(e^\pi - e^{\frac{\pi}{3}}\right)$ [2]

Oct/Nov 2013/11/12/1 (part)

18. The curve C has polar equation $r = 2 \sin \theta (1 - \cos \theta)$, for $0 \leq \theta < \pi$. Find $\frac{dr}{d\theta}$ and hence find the polar coordinates of the point of C that is furthest from the pole. $\left(2 \cos \theta - 2 \cos 2\theta; \left(\frac{3\sqrt{3}}{2}, \frac{2}{3}\pi\right)\right)$ [5]

Sketch C . [2]

Find the exact area of the sector from $\theta = 0$ to $\theta = \frac{1}{4}\pi$. $\left(\frac{5}{16}\pi - \frac{1}{2} - \frac{\sqrt{2}}{3}\right)$ [6]

Oct/Nov 2013/13/10

19. Find the area of the region enclosed by the curve with polar equation $r = 2(1 + \cos \theta)$, for $0 \leq \theta < 2\pi$. (6π) [4]

May/June 2013/11/12/1

20. Use the identity $2 \sin P \cos Q \equiv \sin(P + Q) + \sin(P - Q)$ to show that

$$2 \sin \theta \cos \left(\theta - \frac{1}{4}\pi\right) \equiv \cos \left(2\theta - \frac{3}{4}\pi\right) + \frac{1}{\sqrt{2}}. \quad [3]$$

A curve has polar equation $r = 2 \sin \theta \cos \left(\theta - \frac{1}{4}\pi\right)$, for $0 \leq \theta \leq \frac{3}{4}\pi$. Sketch the curve and state the polar equation of its line of symmetry, justifying your answer. $\left(\theta = \frac{3\pi}{8}\right)$ [3]

Show that the area of the region enclosed by the curve is $\frac{3}{8}(\pi + 1)$. [6]

May/June 2013/13/10

21. Find the cartesian equation corresponding to the polar equation $r = \sqrt{2} \sec \left(\theta - \frac{1}{4}\pi\right)$. $(x + y = 2)$ [3]

Sketch the graph of $r = (\sqrt{2}) \sec \left(\theta - \frac{1}{4}\pi\right)$, for $-\frac{1}{4}\pi \leq \theta \leq \frac{3}{4}\pi$, indicating clearly the polar [2]

coordinates of the intersection with the initial line. **(2, 0)**

Oct/Nov 2012/11/12/1

22. The curve C has polar equation $r = 1 + 2 \cos \theta$. Sketch the curve for $-\frac{2}{3}\pi \leq \theta < \frac{2}{3}\pi$. [2]

Find the area bounded by C and the half-lines $\theta = -\frac{1}{3}\pi, \theta = \frac{1}{3}\pi$. **$(\pi + \frac{5}{2}\sqrt{3})$** [4]

Oct/Nov 2012/13/5

23. The curve C has polar equation $r = 2 + 2 \cos \theta$, for $0 \leq \theta \leq \pi$. Sketch the graph of C . [2]

Find the area of the region R enclosed by C and the initial line. **(3π)** [4]

The half-line $\theta = \frac{1}{5}\pi$ divide R into two parts. Find the area of each part, correct to 3 decimal places. **(4.712, 4.713)** [3]

May/June 2012/11/12/4

24. The curve C has cartesian equation

$$(x^2 + y^2)^2 = a^2(x^2 - y^2),$$

where a is a positive constant. Show that C has polar equation

$$r^2 = a^2 \cos 2\theta. \quad [2]$$

Sketch C for $-\pi < \theta < \pi$. [2]

Find the area of the sector between $\theta = -\frac{1}{4}\pi$ and $\theta = \frac{1}{4}\pi$. **$(\frac{a^2}{2})$** [3]

Find the polar coordinates of all points of C where the tangent is parallel to the initial line. **$(\frac{a}{\sqrt{2}}, \pm \frac{\pi}{6}), (\frac{a}{\sqrt{2}}, \pm \frac{5\pi}{6})$** [7]

May/June 2012/13/11E

25. The curve C has polar equation $r = 3 + 2 \cos \theta$, for $-\pi < \theta \leq \pi$. The straight line l has polar equation $r \cos \theta = 2$. Sketch both C and l on a single diagram. [3]

Find the polar coordinates of the points of intersection of C and l . **$(4, \pm \frac{\pi}{3})$** [4]

The region R is enclosed by C and l , and contain the pole. Find the area of R . **$(\frac{22\pi}{3} - \frac{5\sqrt{3}}{2})$** [6]

Oct/Nov 2011/11/12/10

26. The curve C has polar equation $r = 1 + \sin \theta$ for $-\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi$. Draw a sketch of C . [2]

The area of the region enclosed by the initial line, the half-line $\theta = \frac{1}{2}\pi$, and the part of C for which θ is positive, is denoted by A_1 . The area of the region enclosed by the initial line, and the part of C for which θ is negative, is denoted by A_2 . Find the ratio of $A_1:A_2$, giving your answer correct to 1 decimal place. **(12.2: 1.0)** [8]

Oct/Nov 2011/13/8

27. The curve C has polar equation $r = 2 \cos 2\theta$. Sketch the curve for $0 \leq \theta < 2\pi$. [4]

Find the exact area of one loop of the curve. $\left(\frac{\pi}{2}\right)$ [4]

May/June 2011/11/12/2

28. The curves C_1 and C_2 have polar equations

$$C_1: r = a$$

$$C_2: r = 2a \cos 2\theta, \text{ for } 0 \leq \theta \leq \frac{1}{4}\pi,$$

where a is a positive constant. Sketch C_1 and C_2 on the same diagram. [3]

The curves C_1 and C_2 intersect at the point with polar coordinates (α, β) . State the value of β . [1]
 $\left(\frac{\pi}{6}\right)$

Show that the area of the region bounded by the initial line, the arc of C_1 from $\theta = 0$ and $\theta = \beta$, and the arc of C_2 from $\theta = \beta$ to $\theta = \frac{1}{4}\pi$ is

$$a^2 \left(\frac{1}{6}\pi - \frac{1}{8}\sqrt{3} \right).$$
 [4]

May/June 2011/13/6

29. The curves C_1 and C_2 have polar equations given by

$$C_1: r = 3 \sin \theta, \quad 0 \leq \theta < \pi,$$

$$C_2: r = 1 + \sin \theta, \quad -\pi \leq \theta \leq \pi,$$

- (i) Find the polar coordinates of the points, other than the pole, where C_1 and C_2 meet. [2]

$$\left(\left(\frac{3}{2}, \frac{\pi}{6} \right), \left(\frac{3}{2}, \frac{5\pi}{6} \right) \right)$$

- (ii) In a single diagram, draw sketch graphs of C_1 and C_2 . [3]

- (iii) Show that the area of the region which is inside C_1 but outside C_2 is π . [3]

Oct/Nov 2010/1/8

30. The curve C has polar equation

$$r = a(1 - e^{-\theta}),$$

where a is a positive constant and $0 \leq \theta < 2\pi$.

- (i) Draw a sketch of C . [3]

- (ii) Show that the area of the region bounded by C and the lines $\theta = \ln 2$ and $\theta = \ln 4$ is

$$\frac{1}{2}a^2 \left(\ln 2 - \frac{13}{32} \right).$$
 [4]

May/June 2010/11/12/2

31. The curve C has polar equation

$$r = \frac{a}{1 + \theta},$$

where a is a positive constant and $0 \leq \theta \leq \frac{1}{2}\pi$.

- (i) Show that r decreases as θ increases. [2]
 (ii) The point P of C is further from the initial line than any other point of C . Show that, at P ,

$$\tan \theta = 1 + \theta,$$

and verify that this equation has a root between 1.1 and 1.2. [4]

- (iii) Draw a sketch of C . [3]

- (iv) Find the area of the region bounded by the initial line, the line $\theta = \frac{1}{2}\pi$ and C , leaving your answer in terms of π and a . $\left(\frac{\pi a^2}{2(\pi+2)}\right)$ [3]

May/June 2010/13/11

32. The curve C has polar equation

$$r = a \sin 3\theta,$$

where $0 \leq \theta \leq \frac{1}{3}\pi$.

- (i) Show that the area of the region enclosed by C is $\frac{1}{12}\pi a^2$. [3]

- (ii) Show that, at the point of C at maximum distance from the initial line,
 $\tan 3\theta + 3 \tan \theta = 0$. [3]

- (iii) Use the formula

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

to find this maximum distance. $\left(\frac{9a}{16}\right)$ [4]

- (iv) Draw a sketch of C . [2]

Oct/Nov 2009/1/10

33. Draw a sketch of the curve C whose polar equation is $r = \theta$, for $0 \leq \theta \leq \frac{1}{2}\pi$. [2]

On the same diagram draw the line $\theta = \alpha$, where $0 < \alpha < \frac{1}{2}\pi$. [1]

The region bounded by C and the line $\theta = \frac{1}{2}\pi$ is denoted by R . Find the exact value of α for which the line $\theta = \alpha$ divides R into two regions of equal area. $\left(\pi(16)^{-\frac{1}{3}}\right)$ [4]

May/June 2009/1/5

34. The curve C has polar equation

$$r = \left(\frac{1}{2}\pi - \theta\right)^2.$$

where $0 \leq \theta \leq \frac{1}{2}\pi$. Draw a sketch of C . [3]

Find the area of the region bounded by C and the initial line, leaving your answer in terms of π . $\left(\frac{\pi^5}{320}\right)$ [3]

Oct/Nov 2008/1/3

35. The curves C_1 and C_2 have polar equations

$$r = \theta + 2 \quad \text{and} \quad r = \theta^2$$

respectively, where $0 \leq \theta \leq \pi$.

(i) Find the polar coordinates of the point of intersection of C_1 and C_2 . $((4, 2))$ [2]

(ii) Sketch C_1 and C_2 on the same diagram. [2]

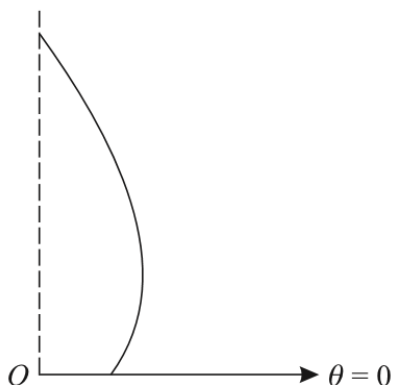
(iii) Find the area bounded by C_1 , C_2 and the line $\theta = 0$. $\left(\frac{92}{15}\right)$ [3]

May/June 2008/1/4

36. The curve C has polar equation $r = \theta \sin \theta$, where $0 \leq \theta \leq \pi$. Draw a sketch of C . [2]

Find the area of the region enclosed by C , leaving your answer in terms of π . $\left(\frac{1}{12}\pi^3 - \frac{1}{8}\pi\right)$ [7]
Oct/Nov 2007/1/7

37.



The diagram shows the curve C with polar equation $r = e^\theta$, where $0 \leq \theta \leq \frac{1}{2}\pi$. Find the maximum distance of a point of C from the line $\theta = \frac{1}{2}\pi$, giving the answer in exact form. $\left(\frac{\sqrt{2}}{2}e^{\frac{\pi}{4}}\right)$ [4]

Find the area of the finite region bounded by C and the lines $\theta = 0$ and $\theta = \frac{1}{2}\pi$, giving the answer in exact form. $\left(\frac{1}{4}(e^\pi - 1)\right)$ [3]

May/June 2007/1/5

38. Draw a diagram to illustrate the region R which is bounded by the curve whose polar equation is $r = \cos 2\theta$ and the lines $\theta = 0$ and $\theta = \frac{1}{6}\pi$. [2]

Determine the exact area of R . $\left(\frac{\pi}{24} + \frac{\sqrt{3}}{32}\right)$ [4]

May/June 2006/1/2

39. The curve C has equation

$$r = 10 \ln(1 + \theta),$$

where $0 \leq \theta \leq \frac{1}{2}\pi$. Draw a sketch of C . [2]

Use the substitution $w = \ln(1 + \theta)$ to show that the area of the sector bounded by the line $\theta = \frac{1}{2}\pi$ and the arc of C joining the origin to the point where $\theta = \frac{1}{2}\pi$ is

$$50(b^2 - 2b + 2)e^b - 100,$$

where $b = \ln\left(1 + \frac{1}{2}\pi\right)$. [6]

Oct/Nov 2006/1/7

40. The curves C_1 and C_2 have polar equations

$$r = 4 \cos \theta \quad \text{and} \quad r = 1 + \cos \theta$$

where $-\frac{1}{2}\pi \leq \theta \leq \frac{1}{2}\pi$.

- (i) Show that C_1 and C_2 meet at the points $A\left(\frac{4}{3}, \alpha\right)$ and $B\left(\frac{4}{3}, -\alpha\right)$, where α is the acute angle such that $\cos \alpha = \frac{1}{3}$. [2]
- (ii) In a single diagram, draw sketch graphs of C_1 and C_2 . [3]
- (iii) Show that the area of the region bounded by the arcs OA and OB of C_1 , and the arc AB of C_2 , is

$$4\pi - \frac{1}{3}\sqrt{2} - \frac{13}{2}\alpha. \quad [7]$$

Oct/Nov 2005/1/12O

41. The curve C has polar equation

$$r = \frac{\pi - \theta}{\theta},$$

where $\frac{1}{2}\pi \leq \theta \leq \pi$.

- (i) Draw a sketch of C . [3]
- (ii) Show that the area of the region bounded by the line $\theta = \frac{1}{2}\pi$ and C is

$$\pi\left(\frac{3}{4} - \ln 2\right). \quad [5]$$

May/June 2005/1/6

42. The curve C has polar equation

$$r = e^{\frac{1}{5}\theta}, \quad 0 \leq \theta \leq \frac{3}{2}\pi.$$

- (i) Draw a sketch of C .

[2]

Oct/Nov 2004/1/4 (part)

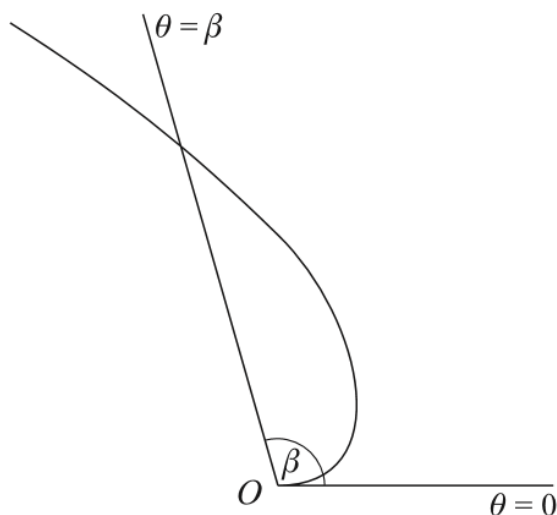
43. The curve C has equation

$$(x^2 + y^2)^2 = 4xy.$$

- (i) Show that the polar equation of C is $r^2 = 2 \sin 2\theta$. [2]
 (ii) Draw a sketch of C , indicating any lines of symmetry as well as the form of C at the pole. $\left(\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}\right)$ [5]
 (iii) Write down the maximum possible distance of a point of C from the pole. $(\sqrt{2})$ [1]

May/June 2004/1/3

- 44.



The curve C has polar equation

$$r = \theta^{\frac{1}{2}} e^{\frac{\theta^2}{\pi}},$$

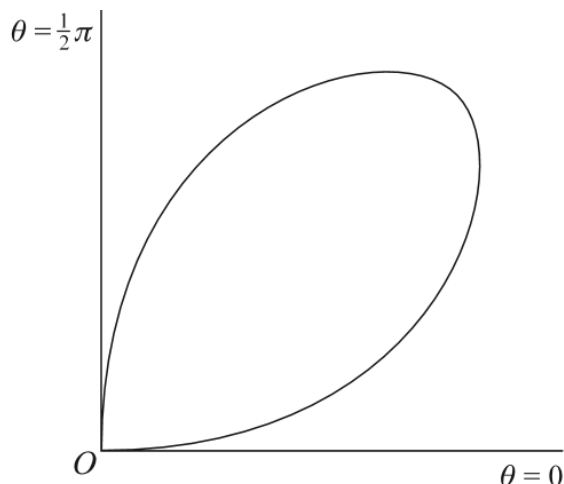
where $0 \leq \theta \leq \pi$. The area of the finite bounded by C and the line $\theta = \beta$ is π (see diagram). Show that

$$\beta = (\pi \ln 3)^{\frac{1}{2}}.$$

[6]

Oct/Nov 2013/1/1

45.



The diagram shows one loop of the curve whose polar equation is $r = a \sin 2\theta$, where a is a positive constant. Find the area of the loop, giving your answer in terms of a and π . $\left(\frac{\pi a^2}{8}\right)$ [4]

May/June 2003/1/1

46. The curve C has polar equation $r\theta = 1$, for $0 < \theta \leq 2\pi$.

(i) Use the fact that $\frac{\sin \theta}{\theta}$ tends to 1 as θ tends to 0 to show that the line with cartesian equation $y = 1$ is an asymptote to C . [2]

(ii) Sketch C . [1]

The points P and Q on C correspond to $\theta = \frac{1}{6}\pi$ and $\theta = \frac{1}{3}\pi$ respectively.

(iii) Find the area of the sector OPQ , where O is the origin. $\left(\frac{3}{2\pi}\right)$ [3]

Oct/Nov 2002/1/5

47. The curve C has polar equation

$$r = a(1 + \cos \theta), \quad -\pi < \theta \leq \pi,$$

where a is a positive constant.

(i) Sketch C . [2]

(ii) Show that the area of the region enclosed by C is $\frac{3}{2}\pi a^2$. [5]

(iii) The point on C with polar coordinates (r, θ) has cartesian coordinates (x, y) . Find the minimum value of y . $\left(-\frac{3\sqrt{3}}{a}a\right)$ [6]

May/June 2002/1/12O